

Let $(X_n)_{n \geq 1}$ be a sequence of non-negative integer RVs.

Example: $X_n = \#$ of triangles in $G_{n,p}$, $p_i = p(n)$.

Recall: First moment method

If $\mathbb{E}X_n \rightarrow 0$, then $\mathbb{P}(X_n > 0) \rightarrow 0$.

Pf: By Markov's Inequality: $\mathbb{P}(X_n \geq 1) \leq \mathbb{E}X_n / 1 = \mathbb{E}X_n$.

Ex: If $p = p(n) = o(n^{-1})$ and $X_n = \#$ triangles in $G_{n,p}$

$$\mathbb{E}X_n = \binom{n}{3} p^3 \sim \frac{n^3}{6} p^3 = o(n^3 n^{-3}) \rightarrow 0.$$

So, $\mathbb{P}(\underbrace{G_{n,p} \text{ contains a triangle}}_{X_n > 0}) \rightarrow 0$

"with high probability" whp } $G \sim G_{n,p}$
 "asymptotically almost surely" aas } contains no triangles

Q: But what if we want the opposite?

For some $p = p(n)$, $G \sim G_{n,p}$, $\mathbb{P}(\underbrace{G \text{ contains a } \Delta}_{X_n > 0}) \rightarrow 1$?

$\rightarrow$ Is $\mathbb{E}X_n \rightarrow \infty$ enough?

Ex: $X_n = \begin{cases} 0 & \text{wp } 1 - 1/n \\ n^2 & \text{wp } 1/n \end{cases} \Rightarrow \mathbb{E}X_n = n^2 \cdot 1/n = n \rightarrow \infty$
 but $\mathbb{P}(X_n > 0) \rightarrow 0$.

Recall: (Variance) $\text{Var}(X) = \mathbb{E}[(X - \mathbb{E}X)^2] = \mathbb{E}X^2 - (\mathbb{E}X)^2$

Chebyshev's Inequality: $\mathbb{P}(|X - \mathbb{E}X| \geq t) \leq \frac{\text{Var}(X)}{t^2}, \exists t > 0$

Proof: Apply Markov's Inequality to $Y = (X - \mathbb{E}X)^2$.

Corollary: For any non-negative RV X ,

$$P(X=0) \leq \frac{\text{Var}(X)}{(\mathbb{E}X)^2}.$$

Second moment method (version 1)

If $(X_n)_{n \geq 1}$ are non-negative RVs with $\mu_n = \mathbb{E}X_n$,

and $\text{Var}(X_n) = o(\mu_n^2)$, then

$$P(X_n = 0) \rightarrow 0.$$

Ex: $p = p(n) \gg n^{-1}$ $G \sim G_{n,p}$ $X_n = \# \text{ triangles } \Delta$

$$\bullet \mathbb{E}X_n = \binom{n}{3} p^3 \rightarrow \infty$$

$$\bullet \text{Var}(X_n) = \underbrace{\mathbb{E}[X_n^2]} - (\mathbb{E}X_n)^2$$

↳ How to calculate this?

$$\rightarrow X_n = \sum_{\substack{\{a,b,c\} \subset V \\ \text{distinct}}} \mathbb{1}_{A_{a,b,c}} \quad ; \quad A_{a,b,c}: \text{triangle b/w vertices } a,b,c$$

$$\mathbb{E}X_n \stackrel{\text{linearity}}{=} \sum_{\substack{\{a,b,c\} \subset V \\ \text{distinct}}} \overbrace{P(A_{a,b,c})}^{=p^3} = \binom{n}{3} p^3 \sim \frac{n^3}{6} p^3.$$

$$\mathbb{E}X_n^2 = \mathbb{E} \left[\sum_{a,b,c} \mathbb{1}_{A_{a,b,c}} \sum_{a',b',c'} \mathbb{1}_{A_{a',b',c'}} \right]$$

$$= \sum_{a,b,c} \sum_{a',b',c'} \mathbb{E} [\mathbb{1}_{A_{a,b,c}} \mathbb{1}_{A_{a',b',c'}}]$$

$$= \sum_{a,b,c} \sum_{a',b',c'} P(A_{a,b,c} \cap A_{a',b',c'}).$$

Generally: if $X = I_1 + \dots + I_k$ where I_i are indicators of some events A_i (X "counts" the events) then

$$\mathbb{E}[X] = \sum_i \mathbb{P}(A_i), \quad \mathbb{E}[X^2] = \sum_{i,j} \mathbb{P}(A_i \cap A_j)$$

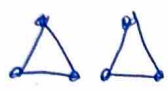
(3)

Back to Δ s:

$$\mathbb{E}[X_n^2] = \sum_{\substack{S \subseteq V \\ |S|=3}} \sum_{\substack{T \subseteq V \\ |T|=3}} \mathbb{P}(A_S \cap A_T)$$

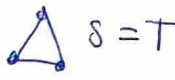
$\uparrow$
 S spans triangle

Overlap of S and T :

• $|S \cap T| = 0$:  $\binom{n}{3} \binom{n-3}{3} p^6 \sim \frac{n^6}{36} p^6 \sim \mu^2$

• $|S \cap T| = 1$:  $\binom{n}{3} \cdot 3 \binom{n-3}{2} p^6 \sim O(n^5 p^6)$

• $|S \cap T| = 2$:  $\binom{n}{3} 3 \cdot n \cdot p^5 = O(n^4 p^5)$

• $|S \cap T| = 3$:  $S = T$ $\binom{n}{3} p^3 = O(n^3 p^3) = \mu$

$\mu = \mu_n = \Theta(n^3 p^3)$ and $p \gg n^{-1}$, so $\mu \rightarrow \infty$.

• $\frac{n^5 p^6}{\mu^2} = \frac{n^5 p^6}{n^6 p^6} = \frac{1}{n} = o(1)$

• $\frac{n^4 p^5}{\mu^2} = \frac{1}{n^2 p} = o\left(\frac{1}{n^2 \cdot 1/n}\right) = o(1)$

• $\frac{\mu}{\mu^2} = \frac{1}{\mu} = o(1)$

So $\mathbb{E}[X_n^2] \sim \mu^2 \left(\Rightarrow (1+o(1))\mu^2 \right)$
 $\text{Var}(X_n) = \mathbb{E}[X_n^2] - (\mathbb{E}[X_n])^2 = o(\mu^2)$

By second moment method

$\mathbb{P}(\mathcal{G} \text{ contains a } \Delta)$

$= \mathbb{P}(X_n > 0)$

$= 1 - \mathbb{P}(X_n = 0) \rightarrow 1$

Remarks:

1) The condition $\text{Var}(X_n) = o(\mu^2)$ is equivalent to $E[X_n^2] \sim \mu^2$.

2) $E[X_n^2] = \sum_{i,j} P(A_i \cap A_j)$
 $\uparrow$ the same if A_i, A_j independent

$$E[X_n]^2 = \sum_{i,j} P(A_i) P(A_j)$$

Write $i \neq j$ if $i \neq j$ and A_i, A_j are dependent

$$\text{Var}(X_n) = \sum_{i,j} \underbrace{(P(A_i \cap A_j) - P(A_i)P(A_j))}_{=0 \text{ if } A_i, A_j \text{ indep.}}$$

$$= \sum_i (P(A_i) - P(A_i)^2) + \sum_i \sum_{j \neq i} (P(A_i \cap A_j) - P(A_i)P(A_j))$$

$$\leq \mu_n + \sum_i \sum_{j \neq i} P(A_i \cap A_j) \quad (\text{just an UB for } \text{Var}(X_n), \text{ but often enough})$$

Second moment method, V2

$(X_n)_{n \geq 1}$ non-negative rvs, $\mu_n := E[X_n] \rightarrow \infty$ and $\Delta_n := \sum_i \sum_{j \neq i} P(A_i \cap A_j) = o(\mu_n^2)$.

Then $P(X_n > 0) \rightarrow 1$.

Proof: $\text{Var}(X_n) \leq \mu_n + \Delta_n = o(\mu^2)$ as $\mu_n \rightarrow \infty$.

Now apply second moment method, version 1.

(Δ_n) will reappear later in the course when we talk about Janson's ineq.

$$\rightarrow \text{Var}(X_n) = \sum_{i,j} \underbrace{(P(A_i \cap A_j) - P(A_i)P(A_j))}_{=: C_{ij}} \Rightarrow \sum_{i,j} = \underbrace{\sum_{i=j}}_{\substack{\text{keep} \\ \text{but} \\ \text{upper} \\ \text{bound}}} + \underbrace{\sum_i \sum_{j \neq i}}_{\substack{\text{keep} \\ \text{but} \\ \text{UB.}}} + \underbrace{\sum_i \sum_{j \neq i}}_{\substack{A_i, A_j \text{ dep.} \\ A_j, A_i \text{ indep.} \\ \text{for } C_{ij} = 0}}$$

Back to triangles!

Recall from L1:

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Let P be a property of graphs (eg "contains a triangle").

Then a function $p^*(n)$ is a threshold for P

• $\frac{p(n)}{p^*(n)} \rightarrow 0$ implies $P(G_{n,p} \text{ has } P) \rightarrow 0$
"whp P does not hold"

• $\frac{p(n)}{p^*(n)} \rightarrow \infty$ $\Rightarrow$ $P(G_{n,p} \text{ has } P) \rightarrow 1$
" P holds whp"

Note: Every non-trivial monotone graph property has a threshold.
not $\emptyset$ or set of all graphs on n vertices if P holds and we add more edges, P still holds

(Bollobás - Thomason 1987, no proof)

We just proved: $1/n$ is a threshold function for $G_{n,p}$ to contain $\triangle K_3$.

In L1: $n^{-2/3}$ threshold for $G_{n,p}$ to contain $\square K_4$.

Generally? When does $G_{n,p}$ contain a copy of some fixed graph H ?

H : v vertices, e edges

X : # of copies of H in $G_{n,p}$

$E[X] = \frac{\overset{\text{select } v \text{ vertices}}{n(n-1)(n-2)\dots(n-v+1)}}{\text{Aut}(H)} \cdot p^e = \Theta(n^v p^e)$
 $\uparrow$ automorphisms of H .

$\rightarrow$ suggests threshold is $n^{-v/e}$.

Not always true! For example: $H = \square$ $v=5$ $e=7$

For $p \gg n^{-5/7}$, $E[X] \rightarrow \infty$. But $n^{-5/7} < n^{-2/3}$ threshold for K_4

So for $n^{-5/7} \ll p(n) \ll n^{-2/3}$, $E[X] \rightarrow \infty$ but whp, no $\square$ exists in $G_{n,p}$

so no $H = \square$ exists, neither.

What went wrong? H has a denser subgraph $K_4 = \square$ which is the "bottleneck".

Def: • Edge density of a graph G with v vertices, e edges is $d(G) = e/v$.

• G is called balanced if for every subgraph G' , $d(G') \leq d(G)$.

Theorem (no proof) If H is balanced with v vertices and e edges, then

$$p^*(n) = n^{-v/e} = n^{-1/d(H)}.$$

is a threshold function for $G_{n,p}$ to contain a copy of H .

→ Proof is by second moment method, similar to Δ and $\square$.

→ Non-balanced H : Threshold is $n^{-1/d(H')}$, where H' is the densest subgraph of H .

Ex 1 3-AP in random subset of $[n] := \{1, \dots, n\}$.

Let $S_{n,p}$ be a random subset of $[n]$, each element included w.p.

Proposition $p^*(n) = n^{-2/3}$ is a threshold for the appearance of an AP of length 3 (3-AP).

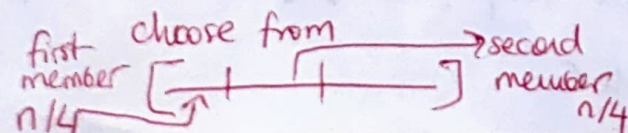
[3AP: a seq. of 3# (a, b, c) s.t. $b - c = c - b \Rightarrow a + c = 2b$]

Observation: If we know two elements of a 3AP, there are only three possibilities for the third one.

$$\mathbb{E}[X] = \Theta(n^2 p^3)$$

lower bound.

choose 2 members, third is given



• If $\frac{p(n)}{n^{-2/3}} \rightarrow 0$ then $\mathbb{E}X = \Theta(n^2 p^3) \rightarrow 0$ so whp $\lambda = 0$. ✓

(7)

• $\frac{p(n)}{p^+(n)} \rightarrow \infty$. Then $\mu = \mathbb{E}X \rightarrow \infty$

$$\Delta_n = \sum_{I \text{ 3-AP}} \sum_{\substack{J \text{ 3-AP} \\ J \sim I}} \mathbb{P}(A_I \cap A_J)$$

where we let A_I be the event that the 3AP I is present.

• Q: When $I \sim J$? If I and J share at least one element (and $I \neq J$).

(1) • $|I \cap J| = 1$:



→ $\Theta(n^2)$ possibilities for J

→ 3 " for common element

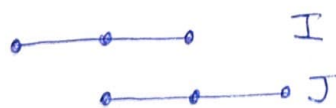
→ Fix one more element of I (n choices)

→ 3 possibilities for final element

Total: $\Theta(n^3)$ choices for I, J $\mathbb{P}(A_I \cap A_J) = p^5$

Contribution: $\Theta(n^3 p^5)$ to Δ_n .

(2) • $|I \cap J| = 2$:



$$\mathbb{P}(A_I \cap A_J) = p^4$$

→ $\Theta(n^2)$ choices for I

→ $\binom{3}{2}$ " " 2 common elements

→ 3 " " final element of J

Contribution $\Theta(n^2 p^4)$ to Δ_n .

$$\Delta_n = O(n^3 p^5) + O(n^2 p^4)$$

$$\frac{\Delta_n}{\mu^2} = \frac{O(n^3 p^5 + n^2 p^4)}{O(n^4 p^6)} = O\left(\frac{1}{np} + \frac{1}{n^2 p^2}\right) \rightarrow 0 \text{ as } p \gg n^{-2/3}$$

So, $\mu_n \rightarrow 0$, $\Delta_n = o(\mu_n^2)$ so $P(X > 0) \rightarrow 1$ by SMH. $\square$

Second moment method v3 (no proof & non-examinable)

Sometimes $P(X > 0) \rightarrow 0$, but we can still use a lower bound on how quickly!

Paley-Zigund inequality

For any non-negative rv X : $P(X > 0) \geq \frac{E[X]^2}{E[X^2]}$

• If $E[X^2] \sim \mu^2 = E[X]^2$: shows $P(X > 0) \rightarrow 1$ like normal SMH.

• If $E[X^2] > \mu^2$: at least gives a lower bound for $P(X > 0)$.